

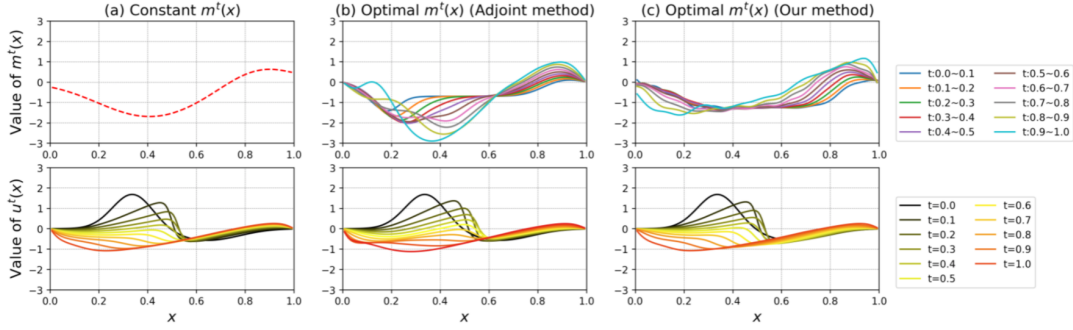


Figure 5: The external force and trajectories with the external force using Burgers' equation. The three columns are (a) a constant external force, (b) an optimal external force using the adjoint method, (c) an optimal external force using our method. The first row shows the time-discretized optimal control inputs m^t ($t = 0, \Delta t, \dots, (n-1)\Delta t$). The second row shows the time evolution of Burgers' equation with the external force in the first row.

$m(t, x)$ reads as

$$\begin{cases} \frac{\partial u}{\partial t} = -u \cdot \frac{\partial u}{\partial x} + \nu \frac{\partial^2 u}{\partial x^2} + m(t, x) & \text{in } [0, T] \times \Omega \\ u(t, x) = 0 & \text{on } [0, T] \times \partial\Omega \\ u(0, x) = u^0(x) \end{cases} \quad (18)$$

where ν is a viscosity parameter, and $u_0(t)$ is an initial condition. During the experiment, we consider the control problem minimizing

$$\min_{u, m} \frac{1}{2} \int_{\Omega} |u(T, x) - u_d(x)|^2 dx + \frac{\alpha}{2} \int_{\Omega \times [0, T]} |m(t, x)|^2 dx dt \quad (19)$$

subject to Burgers' equation (18), given a target $u_d(x)$. In this case, we control an external force m^t . We set $\alpha = 0.01$, $\Omega = [0, 1]$, $T = 1$, and $\Delta t = 0.1$. Also, we set the viscosity parameter $\nu = 0.01$ to generate a shock wave.

The solution operator is well trained during Phase 2 with a relative error 0.0020 for test data. Figure 5 and Table 2 show the result of our method compared to the adjoint method for optimal control (Phase 2). A time step size for the adjoint method is chosen as $\Delta t = 0.01$ since the method does not converge when the time step size is set to the coarse grid ($\Delta t = 0.1$) under our setting. In Figure 5, the trajectory in the second and third columns are scattered with less force than the constant external force in the first column. A quantitative comparison of our method to the adjoint method is shown in Table 2. Our framework takes less time compared to the adjoint method, while the objective function values are similar. Our surrogate model can mimic the time evolution of Burgers' equation in the coarse time grid. It makes our method infer faster than the adjoint method for the control optimization problem of time-dependent PDEs.

Conclusion

We presented a general framework for solving PDE-constrained optimal control problems. We designed the surrogate models for PDE solution operators with a reconstruction structure. It allowed our model to solve the optimal

control efficiently by adopting the reconstruction loss as a regularizer. The experimental results demonstrated that the proposed method has a significant gain in time complexity compared to the adjoint method. Also, our framework can be applied flexibly for both data-driven and data-free control problems.

Although our proposed method can achieve many benefits, it cannot completely replace the existing numerical methods. We believe that the two approaches can be complementary. The numerical method has an advantage in terms of accuracy, and our method is computationally efficient. In general, numerical methods slow down exponentially as the number of dimensions increases. We believe that a deep learning method can alleviate this issue.

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